Mira Flynn

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EXPERIENCE

Telemetry and Software Engineer 2 June 2024 – Present
United Launch Alliance Denver, CO

- Maintained critical hardware and software systems for flight telemetry recording and processing, ensuring seamless data acquisition during launch operations
- Supported launches by ensuring error-free performance of data recording systems and processing raw telemetry data into usable formats for post-launch analysis
- Diagnosed and resolved rare and previously unseen telemetry data anomalies during launch operations using internal tools
- Facilitated internal testing by recording test data and collaborating with cross-functional teams to define and deliver required data products
- Collaborated with team members to design and implement process improvements, enabling the team to support an increased launch cadence
- Accelerated launch data processing by enhancing existing tools with new features and improved resiliency, reducing processing time and errors
- Designed new data processing tools to improve onboard video quality and accelerate release times by up to 10x, cutting processing durations from days to under 2 hours
- Enhanced logging functionality in software tools using an internal logging framework to improve traceability and compliance with organizational standards
- Refactored legacy code by translating it from Perl to Python, improving code readability, maintainability, and documentation
- Contributed to low-level decommutator (decom) modernization and improvement efforts in an agile development structure
- Documented all telemetry data received during launches, including detailed metadata, to ensure traceability and adherence to contract requirements
- Adhered strictly to internal procedures to safeguard confidential and controlled information, ensuring compliance with all organizational policies
- Contributed to low-level decommutator (decom) modernization and improvement efforts

Data Analytics and Management Intern June 2022 – August 2022
Pfizer

- Determined which characteristics of the projects within the division were important for the division's leadership
- Designed and programmed an algorithm to process and visualize large amounts of project management data
- Developed a dashboard using Spotfire and Python to highlight the important characteristics I identified

Teaching Assistant- Data Science, Astronomy and Statistics September 2021 – May 2023
Olin College of Engineering Needham, MA

- Assisted in teaching probability and statistics, programming in R and Python, and effective visual communication of information
- Clarified probability and statistics concepts to students outside of the classroom environment
- Advised students when debugging code for assignments and projects
- Assessed student assignments for correctness, clear scientific writing, and effective visual communication of information

Researcher - Shared Mobility Systems June 2020 – August 2020
Olin College of Engineering Needham, MA

- Researched equity in shared mobility systems for the City of Providence alongside professor Alice Paul
- Implemented a model of system demand and usage patterns in R using linear regression, decision trees, and multiple clustering algorithms
- Designed an application to visualize and explore our model using Shiny and Leaflet packages

EDUCATION

Olin College of Engineering August 2019 – May 2023
Bachelor of Science in Engineering: Computing Needham, MA
Recipient of 50% Tuition Merit Scholarship

Selected Coursework: SCOPE (Senior Capstone Program in Engineering), Technology, Accessibility, and Design, Probabilistic Modeling, Astronomy and Statistics, Principles of Integrated Engineering, Discrete Math, Computer Architecture, Complexity Science, Software Systems, Engineering Systems Analysis, Quantitative Engineering Analysis, Foundations of Computer Science, Data Science, Microelectronic Circuits, Modeling and Simulation

TECHNICAL SKILLS

Programming Languages: Python, C++, Perl, R, SQL, Arduino, MATLAB, JavaScript, HTML, PHP, OCAML, C, RobotC, Verilog, Assembly
Developer Tools: Git/GitHub/Bitbucket/GitLab, Jira, Confleunce, VS Code, Linux/Ubuntu/CentOS/RHEL, Amazon Web Services (AWS), RStudio, Spotfire, Alteryx, Jira, macOS, Windows
Libraries: NumPy, Pandas, GeoPandas, SciPy, Shapely, Matplotlib, Flask, Tidyverse, R Shiny, Leaflet, Tensorflow, Keras, PyTorch, Simulink
Telemetry Skills: IRIG 106 Ch4, IRIG 106 Ch10, IRIG 200, TMoIP
General Skills: Computer Science, Software Design, Software Development, Data Processing, Documentation, Legacy Code Maintennance, Automation, Data Science, Back End Programming, Full Stack Programming, Relational Databases, Using APIs, Version Control, Machine Learning, Artificial Intelligence, Process Improvement, Attention to Detail, Verbal and Written Communication, Collaboration, Creativity, Continuous Learning, Analysis, Problem Solving, Time Management